

Recognition Before Recurrence: A Positive Compiler Quadrilemma for Arithmetic Markov Shifts

Anonymous Authors

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Abstract

We study whether a stationary arithmetic recognizer can acquire an exact prime-only recurrent ledger without changing its symbolic object. The answer is negative for a precisely frozen class: one-sided countable loop-allowed directed graphs with no parallel edges, positive scalar one-step weights, a finite orbit-separating visible alphabet, additive total clock $T(\gamma_p) = \log p$, the original graph-step marker z , and the natural whole-vertex operator on $\ell^2(V)$. A literal atomic ledger forces every recurrent component to be one private simple atom cycle. Shared recurrence creates a mixed primitive root; off-core recognition is invisible to every power trace and, when the whole operator is trace class, to its Fredholm determinant. Finite visible separation forces an infinite subsequence $\ell(p) \geq \log p / (4 \log b)$, so the exact clock leaves uniformly large weights on mutually orthogonal cycles and the whole adjacency is noncompact. First return can be trace class, but it changes $1 - z^{\ell(p)}p^{-s}$ into $1 - zp^{-s}$. An exact audit exhausts all 66,066 loop-allowed digraphs on at most four vertices and verifies 844,544 repaired mixed-root certificates with no true failure. It also preserves 18,272 counterexamples to a stronger preregistered connector normal form. We claim only this source-locked synthesis; computation in symbolic systems, period-set realization, renewal zeta formulae, coding inequalities, and trace-ideal boundaries are established prior art.

Research status. The theorem is a scoped incompatibility result for a frozen class of one-sided positive scalar compilers, not a universal period-set theorem. Its four conclusions are recurrent mixed-word closure, exact pruning of off-core recognition, finite-code clock noncompactness, and a changed marker under first return. The exact audit preserves a preregistered connector claim that failed and verifies the repaired statement. The strict tuple is (AO_STRUCTURAL_ARITHMETIC_RELATION, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL). Hence ROUTE_A_REJECTED; Route B is locked.

1 Introduction

Computational dynamics can encode reachability, halting, and periodicity. That broad connection is classical, from generalized-shift models of Turing machines (Kůrka, 1997; Delvenne et al., 2006) to recent Koopman formulations in which absorbing halting states and cycles have direct spectral signatures (Caravelli and Delvenne, 2026). Period sets are also highly realizable: multidimensional shifts can encode complexity classes (Jeandel and Vanier, 2015), while finite and finitely presented systems obey sharp least-period classifications (Doering and Pavlov, 2019; de Jong, 2026). Thus the unrestricted claim that recognition cannot become recurrence is false.

Our question is narrower. Suppose a single one-sided countable Markov graph must expose arithmetic recognition, carry a positive scalar edge roof, retain the original graph-step marker, and own an ordinary Fredholm determinant on the natural vertex Hilbert space. Can its primitive periodic ledger contain exactly one orbit for each prime and no mixed orbit? The object is frozen before the answer is known; a terminal prime projector, a supplied prime alphabet, and an induced time coordinate are disallowed as same-object repairs.

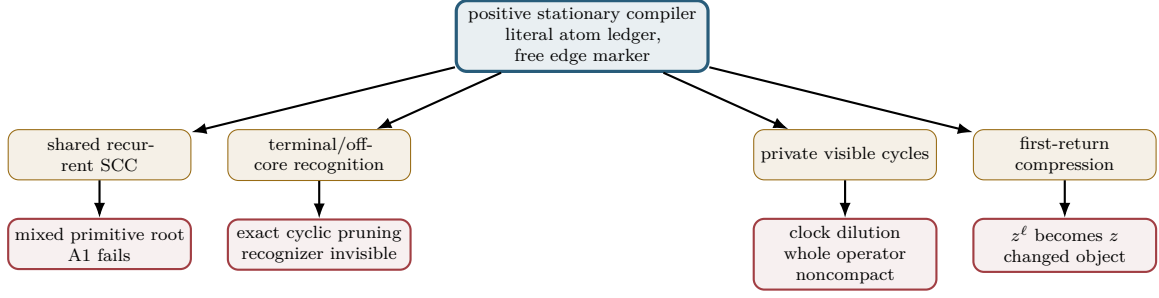


Figure 1: The recognition-to-recurrence quadrilemma. Exhaustiveness is asserted only for the positive scalar compiler class frozen in [Section 3](#).

The answer is a four-way incompatibility, summarized in [Figure 1](#). Shared recurrent computation produces a mixed primitive root. Recognition outside the recurrent core prunes exactly from traces and determinants. Private visible cycles require logarithmic code length; with total roof $\log p$ their whole adjacency is noncompact. First return repairs the operator only by replacing the raw marker $z^{\ell(p)}$ with one return marker z .

The contribution is a quantifier-clean ownership theorem, not a new primitive-root theorem, Kraft inequality, or compactness criterion. Its value is diagnostic: the same source, marker, periodic ledger, and operator cannot be silently exchanged while credit is retained. The exact audit is part of that diagnosis. A stronger preregistered connector normal form fails on 18,272 finite instances; the corrected proof needs only arbitrary mutual paths inside a strongly connected component.

The remainder of the paper separates prior art from the source lock, proves the quadrilemma, reports the exact audit, and closes the positive Route-A candidate. The next benchmark is the arithmetic $ax + b$ and Bost–Connes setting, but only after a new symbolic source lock; its partition function cannot be imported as periodic-orbit credit.

2 Literature and claim boundary

Finite graph zeta and determinant formulae originate with [Bowen and Lanford \(1970\)](#). Countable loop graphs are organized by first-return series, and their zeta functions exhibit the familiar $1/(1 - f)$ structure ([Boyle et al., 2006](#); [Hong, 2011](#)). Circular-code formulae already capture the combinatorics of concatenated return words ([Keller, 1991](#)). More strongly, complex local weights on a renewal shift can realize very flexible holomorphic germs ([Sarig, 2004](#)); positivity and source ownership are therefore indispensable here.

The finite-code estimate used below is elementary distinct-word counting. The classical inequality of [McMillan \(1956\)](#) gives a stronger bound when unique decipherability is assumed, and constrained versions are known for sofic codes ([Béal and Perrin, 2005](#)). We do not claim a new coding inequality. Likewise, ordinary Fredholm determinants require trace-ideal control ([Simon, 2005](#)). Infinite weighted graphs can possess other valid determinant frameworks under different total-weight hypotheses ([Deitmar, 2015](#)); our noncompactness conclusion concerns the specified whole vertex adjacency, not every possible graph operator.

Inducing is also legitimate mathematics. State expansion and time change have classical flow-equivalence lineage ([Parry and Sullivan, 1975](#)). Our marker statement is an ownership firewall: one first-return step is not one edge step. It does not declare induced systems invalid.

The closest current collision is the 2026 Koopman treatment of symbolic computation by [Caravelli and Delvenne \(2026\)](#); the closest period-spectrum collision is the 2026 classification and realization result of [de Jong \(2026\)](#). A bounded primary-source search completed on 15 August 2026 found no statement of the exact conjunction used here. This is not a priority claim. Every component has precedent, and the novelty asserted is only the source-locked conjunction

and its adversarial audit.

Finally, the Bost–Connes system has partition function $\zeta(\beta)$ (Bost and Connes, 1995), and the $ax + b$ semigroup has a canonical C^* -algebraic treatment (Cuntz, 2008). These are benchmarks for the next paper, not instances of the compiler class below.

3 Frozen compiler class

Definition 3.1 (phase space and marker). Let $G = (V, E)$ be one fixed countable loop-allowed directed graph with no parallel edges, generated by a cutoff-independent rule. Its phase space is the one-sided edge shift

$$X_G^+ = \{e_0 e_1 \dots : t(e_j) = o(e_{j+1})\}.$$

A primitive orbit is a nonempty directed closed edge word modulo cyclic rotation that is not a proper temporal power. Reflection is not identified. The free variable z records one edge of this original graph.

Let $\mathcal{A}$ be a countably infinite atom set with norm $N : \mathcal{A} \rightarrow (1, \infty)$. We require multiplicative freeness:

$$\prod_{a \in \mathcal{A}} N(a)^{m_a} = 1, \quad m_a \in \mathbb{Z} \text{ finitely supported} \implies m_a = 0 \quad \forall a. \quad (1)$$

The prime specialization is $\mathcal{A} = \mathbb{P}$ and $N(p) = p$. Enumerate the atoms in nondecreasing norm and assume eventually

$$N(a_j) \leq j^\kappa. \quad (2)$$

For rational primes one may take $\kappa = 2$.

Fix a finite visible alphabet $\mathcal{B}$, $b = |\mathcal{B}| \geq 2$, and after a higher-block recoding a one-edge label map $\lambda : E \rightarrow \mathcal{B}$. If $\Lambda(\gamma)$ is the cyclic label word of an orbit, visibility means

$$\Lambda(\gamma_a) = \Lambda(\gamma_c) \implies a = c. \quad (3)$$

Without (3), an arbitrary inventory may be hidden in countable vertex names.

The nonnegative edge roof $\tau : E \rightarrow [0, \infty)$ is frozen before evaluation. For real $\sigma > 0$ the whole weighted adjacency on $\mathcal{H} = \ell^2(V)$ is

$$L_\sigma \delta_u = \sum_{e: u \rightarrow v} e^{-\sigma \tau(e)} \delta_v. \quad (4)$$

If it is unbounded, the whole-operator gate has already failed. A weighted anisotropic space or an induced return space is a different owned object.

Definition 3.2 (literal atomic ledger). There is exactly one primitive orbit γ_a for every $a \in \mathcal{A}$, there are no other primitive orbits, and

$$T(\gamma_a) := \sum_{e \in \gamma_a} \tau(e) = \log N(a). \quad (5)$$

An extra mixed primitive word is a failure before any cancellation is considered. Temporal repetitions of γ_a are allowed and are not new primitive orbits.

The central connected ledger is

$$\mathcal{L}_G(s, z) = \sum_{[\gamma]} \sum_{\text{primitive } r \geq 1} \frac{z^{r|\gamma|}}{r} e^{-srT(\gamma)}. \quad (6)$$

We first read (6) literally. Only after trace-class ownership is proved may it be identified with $-\log \det(I - zL_s)$.

Recognition edges may be transient, periodically active, or shared recurrent. A terminal support projector, a supplied prime table, a cutoff-dependent alphabet, target-zero data, and an after-the-fact orbit weight are forbidden. Ordinary nondeterministic paths are counted with multiplicity; Boolean path existence is not the scalar semantics of (4).

4 The positive compiler quadrilemma

Theorem 4.1 (recognition-to-recurrence quadrilemma). *Assume the source lock of [Section 3](#). Write $\ell(a) = |\gamma_a|$. Then:*

- (I) *Every γ_a is a simple directed cycle, distinct atom cycles are vertex-disjoint, and every recurrent strongly connected component is exactly one atom cycle. Shared recurrent computation creates an extra mixed primitive root.*
- (II) *Let C_σ retain only the atom-cycle edges on the same vertex space. If $L_\sigma \in \mathcal{S}_1$, then $C_\sigma \in \mathcal{S}_1$ and*

$$\text{Tr } L_\sigma^r = \text{Tr } C_\sigma^r \quad (r \geq 1), \quad (7)$$

so for all $z \in \mathbb{C}$,

$$\det(I - zL_\sigma) = \det(I - zC_\sigma) = \prod_{a \in \mathcal{A}} (1 - z^{\ell(a)} N(a)^{-\sigma}). \quad (8)$$

All terminal and off-core recognition is determinant-invisible.

- (III) *Under (2) and (3), infinitely many atoms satisfy*

$$\ell(a) \geq \frac{\log N(a)}{2\kappa \log b}. \quad (9)$$

For every $\sigma > 0$, the whole adjacency is therefore either unbounded or bounded and noncompact, hence belongs to no finite Schatten class. For primes,

$$\ell(p) \geq \frac{\log p}{4 \log b}, \quad \max_{e \in \gamma_p} e^{-\sigma \tau(e)} \geq b^{-4\sigma} \quad (10)$$

on an infinite subsequence.

- (IV) *First return to one base point per private cycle gives the diagonal operator $R_s e_a = N(a)^{-s} e_a$ and, in its trace-class half-plane,*

$$\det(I - zR_s) = \prod_a (1 - zN(a)^{-s}). \quad (11)$$

The raw and induced local factors agree at $z = 1$, but their free-marker germs agree only if every $\ell(a) = 1$. Infinite length-one cycles contradict finite-alphabet separation.

Corollary 4.2 (compiler consequence). *No compiler in the frozen class can simultaneously have genuine shared recurrent recognition, a literal prime-only ledger, finite local visibility, the exact total roof $\log p$, a compact whole adjacency on natural $\ell^2(V)$, and the unchanged edge marker. Moving recognition off the cycles gives pruning; serializing it on private visible cycles gives clock dilution; sharing it gives mixed primitives; inducing changes time.*

The theorem is a conjunction of exact statements. It neither asserts that arbitrary countable graphs cannot realize a chosen orbit inventory nor rules out signed, matrix-valued, supertrace, or nonlocal cancellations.

5 Recurrent rigidity and the connector repair

Lemma 5.1 (simple and disjoint atom cycles). *Under the literal ledger and (1), every atom cycle is simple and two distinct atom cycles share no vertex.*

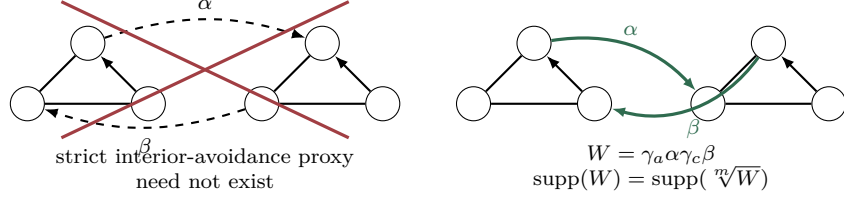


Figure 2: Left: the preregistered strict connector normal form need not exist. Right: arbitrary mutual SCC paths suffice because a word and its primitive root have identical edge support. A one-way connector is not recurrent.

Proof. If distinct cycles meet, rotate their edge words x, y at a shared vertex and write $xy = w^m$ with w primitive. The literal ledger identifies w with some γ_d . Roof additivity gives $N(a)N(c) = N(d)^m$, contradicting (1). If one atom cycle repeats a vertex before closing, split it there into nonempty closed words and take their primitive roots. The same argument yields $N(a) = N(c)^m N(d)^n$, another forbidden exponent relation. $\square$

Lemma 5.2 (mixed root in a shared component). *Every recurrent strongly connected component is exactly one atom cycle.*

Proof. Suppose one component contains distinct atom cycles γ_a and γ_c . They are vertex-disjoint by lemma 5.1. Choose any path α from an attachment vertex of the first cycle to one on the second, and any path β back. The closed word

$$W = \gamma_a \alpha \gamma_c \beta$$

has a primitive root w . Since $W = w^m$, W and w have identical edge support. The ledger identifies w with an atom cycle. Because it meets γ_a , lemma 5.1 forces equality with γ_a ; yet w also contains an edge of γ_c , a contradiction. Every extra edge in a recurrent component lies on a closed walk and yields the same argument. $\square$

No shortest-path or interior-avoidance normal form is used. This distinction is essential. The preregistered finite proxy required both connectors to avoid both cycles internally and failed on 18,272 examples. The repaired statement asks only for mutual reachability, which is exactly what a strongly connected component supplies.

The argument is graph-theoretic and independent of weights. Positivity enters later, when a literal orbit statement is converted into a trace statement without cancellation.

6 Exact pruning and operator ownership

Let V_c be the union of atom-cycle vertices. Permute each cycle forward and fix all other vertices, obtaining a unitary U . If E_{diag} is the diagonal conditional expectation, then

$$D_\sigma = E_{\text{diag}}(U^* L_\sigma), \quad C_\sigma = U D_\sigma.$$

Contractivity of E_{diag} on trace class gives $\|C_\sigma\|_1 \leq \|L_\sigma\|_1$.

Every diagonal coefficient of L_σ^r is a sum over length- r closed walks based at that vertex. Each edge of a closed walk lies in a recurrent component, so lemma 5.2 puts the walk entirely on one atom cycle. Consequently the diagonal coefficients of L_σ^r and C_σ^r coincide. Summing proves (7). The trace-class identity

$$\log \det(I - zL) = - \sum_{r=1}^{\infty} \frac{z^r}{r} \text{Tr } L^r$$

first gives equality near $z = 0$, and entireness extends it to all z . A weighted cycle of length $\ell(a)$ has determinant

$$1 - z^{\ell(a)} \prod_{e \in \gamma_a} e^{-\sigma \tau(e)} = 1 - z^{\ell(a)} N(a)^{-\sigma},$$

which proves (8).

The conclusion is stronger than pruning a finite nilpotent branch: any edge outside the recurrent core is invisible, even if infinitely many such edges feed into or leave cycle vertices. But trace-class ownership is conditional. If the whole operator is not trace class, the formal orbit product is not thereby an ordinary Fredholm determinant.

The terminal control makes the meaning transparent. For

$$L = \begin{pmatrix} 0 & 0 \\ c & x \end{pmatrix} \quad (c, x > 0), \quad \det(I - zL) = 1 - zx.$$

Deleting the feed state changes neither a power trace nor the determinant. A recognizer followed by accept-loop orbitification can therefore implement any decidable support, but the periodic ledger records the terminal output convention rather than recurrent computation.

7 Finite visibility and clock dilution

Let $M_J = \max_{j \leq J} \ell(a_j)$. There are fewer than

$$W_b(L) = \sum_{r=1}^L b^r < \frac{b^{L+1}}{b-1}$$

nonempty words of length at most L , and no more cyclic words. Separation therefore implies $J \leq W_b(M_J)$, hence for all large J ,

$$M_J \geq \frac{\log J}{2 \log b}.$$

At every record index J , $\ell(a_J) = M_J$. Combining this with $\log N(a_J) \leq \kappa \log J$ gives (9) on an infinite subsequence.

Choose on each such cycle an edge e_a of minimum roof. By the exact total clock,

$$\tau(e_a) \leq \frac{\log N(a)}{\ell(a)} \leq 2\kappa \log b, \quad e^{-\sigma \tau(e_a)} \geq b^{-2\kappa \sigma}.$$

The source vertices $u_a = o(e_a)$ lie on distinct private cycles, so δ_{u_a} is orthonormal and weakly null. Positivity yields

$$\|L_\sigma \delta_{u_a}\|_2 \geq b^{-2\kappa \sigma}.$$

A compact operator sends a bounded weakly null sequence to a norm-null sequence. Thus a bounded L_σ is noncompact and cannot belong to any finite Schatten class.

The finite alphabet and orbit separation do real work. If one private two-cycle is supplied per prime, with both roofs $(\log p)/2$, the recurrent direct sum is compact for every $\sigma > 0$ and trace class for $\sigma > 2$. Its factor is $1 - z^2 p^{-\sigma}$, however, and an infinite visible inventory has been hidden in vertex names. A countable one-symbol-per-prime diagonal also escapes, but is exactly a supplied atom inventory.

The proof is elementary; general weighted-shift theory supplies a broader context (Jabłoński et al., 2012). We claim only its consequence for the frozen logarithmic clock.

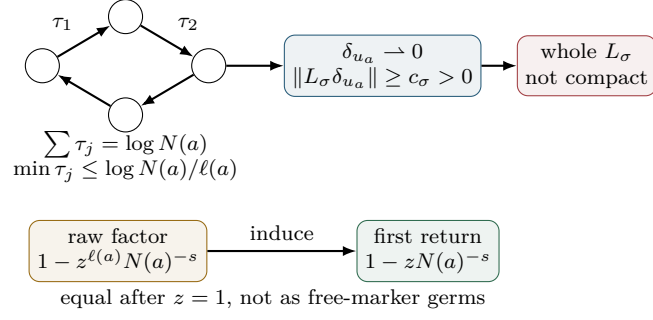


Figure 3: Clock and marker firewall. The exact total roof leaves a uniformly large edge on a logarithmically long private cycle; first return makes one cycle one step and therefore changes $z^{\ell(a)}$ to z .

8 First return and the free-marker firewall

Choose one base vertex on each private cycle. One induced step traverses the whole cycle and has weight

$$\prod_{e \in \gamma_a} e^{-s\tau(e)} = e^{-sT(\gamma_a)} = N(a)^{-s}.$$

Thus $R_s e_a = N(a)^{-s} e_a$, and where $\sum_a N(a)^{-\text{Re}(s)} < \infty$ it is trace class with determinant (11). This is a valid operator.

It is not the original edge-step object. The exact local comparison is

$$\underbrace{1 - z^{\ell(a)} N(a)^{-s}}_{\text{raw graph steps}}, \quad \underbrace{1 - z N(a)^{-s}}_{\text{one return step}}. \quad (12)$$

Setting $z = 1$ forgets the distinction. As free-marker germs they agree only when $\ell(a) = 1$. More generally, if a positive raw determinant equals $\prod_a (1 - z N(a)^{-\sigma})$ at one absolutely convergent real σ , the coefficient of z forces every atom orbit to have length one. A finite alphabet has only finitely many one-letter cyclic words, contradicting infinite separation.

This is a noncommuting ownership square: one may preserve the original marker and lose whole-operator compactness, or induce to a trace-class diagonal and change time. Equality after $z = 1$ does not make the two routes the same object.

9 Exact adversarial audit

The computation was frozen before canonical execution and uses only exact integer, rational, and symbolic arithmetic. Candidate generation and the independent evaluator use different algorithms: Tarjan SCCs versus transitive-closure SCCs, DFS cycles versus vertex-permutation cycles, period-divisor roots versus coordinate-period roots, and Newton identities versus direct recurrent-block reconstruction.

The repaired mixed-root audit totals 844,544/844,544 certificates with no true failure. The failed strict proxy remains in the public artifact ledger: 24 failures on three vertices, 17,952 on four vertices, and 296 among the seeded controls. This is evidence for theorem repair, not a discarded inconvenience.

The terminal graph has 160 states, of which 126 are recurrent and 34 are acyclic decision states. Newton reconstruction equals the product of the 17 recurrent cycle factors. Across eight post-freeze inventories, every proper nonempty support pruning changes the determinant. Signed and matrix controls deliberately lie outside the theorem: a signed three-state nilpotent recurrent matrix has determinant one, and orthogonal matrix branches can annihilate mixed products while retaining pure ones.

Gate	Interpretation	Exact outcome
All loop-allowed digraphs, $n \leq 4$	exhaustive, not sampled	66,066 graphs
Shared-state mixed roots	exhaustive within that census	613,996/613,996
Repaired connector roots	arbitrary mutual SCC paths	161,475/161,475
Seeded $n = 5, \dots, 8$ controls	finite, explicitly nonexhaustive	69,073/69,073
Strict connector proxy	preregistered normal form	18,272 failures
Terminal compiler	126 recurrent, 34 acyclic states	exact pruning
Kraft/clock	12 configurations	6,141 rows, 0 failures
Raw/induced marker	one factor per frozen atom	17/17 differ, agree at $z = 1$

Table 1: Canonical exact evidence. The seeded controls are never described as an exhaustive graph census.

The canonical authority run contains 76/76 passing tests. Two cleared fresh-result runs produced 19/19 byte-identical artifacts with aggregate SHA-256 `ae0aa6d1767bb207d0096df149224995bfb40aba674367a2f300668bfdd88c02`. The final result ledger contains 41 entries; its ledger-file SHA-256 is `6ffbbe5ce1e2a20f0fb00839b89981293c25a8eca2a0995ed384e7448dc7591`. The sealed package additionally checks LF-only text, exact terminal newlines, absence of cache and timestamp fields, Route schema, source separation, and result-set equality.

10 Route closure and conclusion

The exact statements are mathematically positive, but the candidate route is negative. The frozen Route-A tuple is

`(A0_STRUCTURAL_ARITHMETIC_RELATION, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL)`.

A0 receives only structural credit: multiplicative freeness and the logarithmic roof are source-derived. A1 fails because shared recurrence is incompatible with a literal atom-only primitive ledger. A2 fails because the primary whole operator is noncompact under finite visible separation; the induced determinant belongs to a changed marker. A3 and A4 fail because the mechanism is generic for any multiplicatively free atom inventory and does not provide a spectral carrier. Overall:

`ROUTE_A_REJECTED, ROUTE_B_LOCKED.`

The theorem closes a compiler interface, not Symbolic Dynamics as a research family. Its sharp controls identify the open boundaries: signed or matrix-valued trace cancellation, infinite visible alphabets, nonlocal orbit actions, and operator-algebraic constructions require new theorems.

The next paper will test the $\mathbb{N} \rtimes \mathbb{N}^\times$ and Bost–Connes benchmarks without importing their diagonal partition function as a periodic ledger. It must freeze a graph-step marker and ask, before any prime label is read, whether positive semigroup motion is acyclic, whether symmetrization creates universal backtracking or relation cycles, and whether the owned operator is genuinely the same symbolic object. Until those gates are passed, $\zeta(\beta)$ as a partition function is not evidence for a prime-selective recurrent compiler.

References

Marie-Pierre Béal and Dominique Perrin. Codes and sofic constraints. *Theoretical Computer Science*, 340(2):381–393, 2005. doi: 10.1016/j.tcs.2005.03.033.

- Jean-Benoît Bost and Alain Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Mathematica*, 1:411–457, 1995. doi: 10.1007/BF01589495.
- Rufus Bowen and Oscar E. Lanford. Zeta functions of restrictions of the shift transformation. In *Global Analysis*, volume 14 of *Proceedings of Symposia in Pure Mathematics*, pages 43–49. American Mathematical Society, 1970. doi: 10.1090/pspum/014/9985.
- Mike Boyle, Jérôme Buzzi, and Ricardo Gómez. Almost isomorphism for countable state Markov shifts. *Journal für die reine und angewandte Mathematik*, 592:23–47, 2006. doi: 10.1515/CRELLE.2006.021.
- Francesco Caravelli and Jean-Charles Delvenne. Analog and symbolic computation through the Koopman framework. *Journal of Physics: Complexity*, 7:025012, 2026. doi: 10.1088/2632-072X/ae6f5f.
- Joachim Cuntz. C^* -algebras associated with the $ax + b$ -semigroup over $\mathbb{N}$. In *K-Theory and Noncommutative Geometry*, EMS Series of Congress Reports, pages 201–215. European Mathematical Society, 2008. doi: 10.4171/060-1/8.
- Huub de Jong. On sets of periodic orbit lengths in finitely presented dynamical systems. *Ergodic Theory and Dynamical Systems*, pages 1–33, 2026. doi: 10.1017/etds.2026.10313.
- Anton Deitmar. Ihara zeta functions of infinite weighted graphs. *SIAM Journal on Discrete Mathematics*, 29(4):2100–2116, 2015. doi: 10.1137/140957925.
- Jean-Charles Delvenne, Petr Kůrka, and Vincent D. Blondel. Decidability and universality in symbolic dynamical systems. *Fundamenta Informaticae*, 74(4):463–490, 2006. URL <https://arxiv.org/abs/cs/0404021>. Publisher-reported DOI 10.3233/FUN-2006-74406; resolver unavailable on 2026-08-15.
- Madeline Doering and Ronnie Pavlov. A characterization of the sets of periods within shifts of finite type. *Involve*, 12(2):203–220, 2019. doi: 10.2140/involve.2019.12.203.
- Soonjo Hong. The zeta functions of renewal systems. *Ergodic Theory and Dynamical Systems*, 31(3):875–890, 2011. doi: 10.1017/S0143385710000015.
- Zenon Jan Jabłoński, Il Bong Jung, and Jan Stochel. Weighted shifts on directed trees. *Memoirs of the American Mathematical Society*, 216(1017):1–101, 2012. doi: 10.1090/S0065-9266-2011-00644-1.
- Emmanuel Jeandel and Pascal Vanier. Characterizations of periods of multi-dimensional shifts. *Ergodic Theory and Dynamical Systems*, 35(2):431–460, 2015. doi: 10.1017/etds.2013.60.
- Karl Keller. Circular codes, loop counting, and zeta-functions. *Journal of Combinatorial Theory, Series A*, 56(1):75–83, 1991. doi: 10.1016/0097-3165(91)90023-A.
- Petr Kůrka. On topological dynamics of Turing machines. *Theoretical Computer Science*, 174(1–2):203–216, 1997. doi: 10.1016/S0304-3975(96)00025-4.
- Brockway McMillan. Two inequalities implied by unique decipherability. *IRE Transactions on Information Theory*, 2(4):115–116, 1956. doi: 10.1109/TIT.1956.1056818.
- William Parry and Dennis Sullivan. A topological invariant of flows on 1-dimensional spaces. *Topology*, 14(4):297–299, 1975. doi: 10.1016/0040-9383(75)90012-9.
- Omri Sarig. Zeta functions for the renewal shift. RIMS Kôkyûroku 1404, 98–104, 2004. URL <https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/pdf/1404-10.pdf>.
- Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, 2 edition, 2005. doi: 10.1090/surv/120.

A Proof details and analytic continuation boundaries

This appendix records two points that are easy to lose in a compressed presentation.

A.1 Trace-class extraction

The diagonal conditional expectation is contractive on $\mathcal{S}_1$. Therefore the cyclic extraction $C_\sigma = UD_\sigma$ used in [Section 6](#) is trace class whenever L_σ is. No assertion is made when the latter is merely formal. If $s \mapsto L_s$ is trace-norm holomorphic on a connected domain containing a real interval, the extraction is bounded linear and hence trace-norm holomorphic; determinant equality on the real interval then extends by the identity theorem. This does not create continuation beyond the proved trace-class domain.

A.2 Why the primitive root retains both cycles

For a nonempty directed word W , let w be its primitive root, so $W = w^m$. Every occurrence of every edge of W lies in one of the repeated copies of w , and every edge in w occurs in W . Thus $\text{supp}(W) = \text{supp}(w)$. Directed words have no cancellation. Consequently the paths in [lemma 5.2](#) may revisit either atom cycle; no internal-disjointness or shortest-path hypothesis is needed.

A.3 Coefficientwise marker test

At an absolutely convergent real σ , the coefficient of z in the raw connected ledger is

$$\sum_{a:\ell(a)=1} N(a)^{-\sigma},$$

whereas the induced target coefficient is $\sum_a N(a)^{-\sigma}$. Positivity makes equality possible only when the complement is empty. This is why a cancellation-based signed or supertrace model lies outside the theorem.

B Scope declarations and sharp controls

- A disjoint union of one prescribed cycle per desired length realizes an arbitrary primitive inventory. It violates source-visible finite coding or simply hardwires the inventory.
- A one-way connector between two private cycles creates no mixed closed word. Mutual return, not mere attachment, is the hypothesis of [lemma 5.2](#).
- With signed scalar weights, the strongly connected matrix

$$\begin{pmatrix} 0 & u & -u \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

is nilpotent and has determinant one. The underlying mixed primitive words still exist, but their trace contributions cancel.

- Matrix-valued, graded, homological, and supertrace models are not classified. Orthogonal branch products give finite counter-controls to an unscoped scalar conclusion.
- Negative finite edge roofs do not repair the average-roof estimate when the cycle total remains $\log N(a)$; they can instead destroy boundedness. Zero weights destroy the required nonzero orbit product.
- A local or regularized graph zeta may remain meaningful when the whole vertex adjacency is noncompact. Such an object cannot be called its ordinary Fredholm determinant without a separate ownership theorem.

- No target-zero data, zero locations, critical-line fitting, or Route-B continuation is used anywhere in the construction or audit.

The result is therefore a closed theorem for a deliberately narrow positive compiler class and an open invitation at its signed, matrix-valued, and operator-algebraic boundaries.